

Calc 1 – Techniques of Integration

Worksheet A

1. Compute $\int 2x\sqrt{x^2+1} \, dx$ (u -substitution).
2. Compute $\int x \cos x \, dx$ (integration by parts).
3. Compute $\int \sin^3 x \cos^2 x \, dx$.
4. Compute $\int \frac{1}{\sqrt{4-x^2}} \, dx$.
5. Compute $\int \frac{1}{(x-1)(x+2)} \, dx$ (partial fractions).

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Worksheet B

1. Compute $\int_0^1 x e^{x^2} dx$ (u -substitution).
2. Compute $\int \ln x dx$ (integration by parts).
3. Compute $\int \frac{x^2}{\sqrt{1-x^2}} dx$ (trig substitution).
4. Determine whether $\int_1^\infty \frac{1}{x^2} dx$ converges; if so, find its value.
5. Compute $\int \frac{2x+1}{x^2-x-2} dx$ (partial fractions).

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Answer Key (Worksheets A & B)

Worksheet A.

1. $u = x^2 + 1 \Rightarrow du = 2x dx$. Integral $= \int u^{1/2} du = \frac{2}{3}u^{3/2} = \boxed{\frac{2}{3}(x^2 + 1)^{3/2} + C}$.
2. Parts: $u = x$, $dv = \cos x dx \Rightarrow du = dx$, $v = \sin x$. $\int x \cos x dx = x \sin x - \int \sin x dx = \boxed{x \sin x + \cos x + C}$.
3. $\sin^3 x = \sin x(1 - \cos^2 x)$. Let $u = \cos x$, $du = -\sin x dx$. Integral $= -\int (1 - u^2)u^2 du = -\int (u^2 - u^4) du = -\frac{u^3}{3} + \frac{u^5}{5} + C = \boxed{\frac{\cos^5 x}{5} - \frac{\cos^3 x}{3} + C}$.
4. Standard form $\boxed{\arcsin(x/2) + C}$.
5. $\frac{1}{(x-1)(x+2)} = \frac{1/3}{x-1} - \frac{1/3}{x+2}$. Integral $= \boxed{\frac{1}{3} \ln \left| \frac{x-1}{x+2} \right| + C}$.

Worksheet B.

1. $u = x^2$, $du = 2x dx$. $\int_0^1 x e^{x^2} dx = \frac{1}{2} \int_0^1 e^u du = \frac{1}{2}(e - 1) = \boxed{(e - 1)/2}$.
2. Parts: $u = \ln x$, $dv = dx \Rightarrow du = \frac{dx}{x}$, $v = x$. $\int \ln x dx = x \ln x - \int 1 dx = \boxed{x \ln x - x + C}$.
3. $x = \sin \theta$, $dx = \cos \theta d\theta$, $\sqrt{1 - x^2} = \cos \theta$. Integral $= \int \sin^2 \theta d\theta = \frac{1}{2}(\theta - \sin \theta \cos \theta) + C = \boxed{\frac{1}{2} \arcsin x - \frac{1}{2} x \sqrt{1 - x^2} + C}$.
4. $\int_1^b \frac{dx}{x^2} = [-1/x]_1^b = 1 - 1/b \rightarrow \boxed{1}$ as $b \rightarrow \infty$. Converges to 1.
5. Factor: $x^2 - x - 2 = (x - 2)(x + 1)$. $\frac{2x+1}{(x-2)(x+1)} = \frac{A}{x-2} + \frac{B}{x+1}$. Solve: $A = 5/3$, $B = 1/3$. Integral $= \boxed{\frac{5}{3} \ln |x - 2| + \frac{1}{3} \ln |x + 1| + C}$.